Inequalities — problem 7.38

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The vectors $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ with sum $\vec{o}$ are given in a plane. Prove the inequality

$$|\vec{a}| + |\vec{b}| + |\vec{c}| + |\vec{d}| \geq |\vec{a} + \vec{d}| + |\vec{b} + \vec{d}| + |\vec{c} + \vec{d}|.$$

Prove this also for one and three dimensions (AUO 1976).